26 春《电子电路与系统基础（I）》期中考试试题（英文卷）

注意：考试结束时请与答题纸一同上交

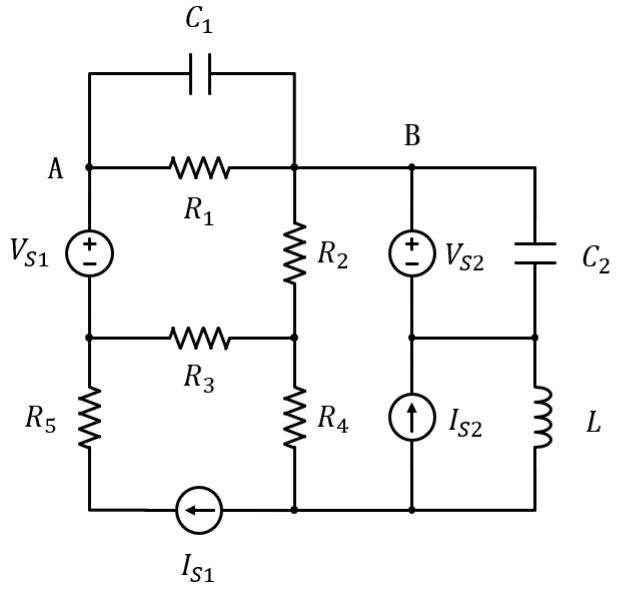
姓名：_____ 学号：_____

注：卷面共三大题，满分为 120 分，以最高 100 分计算成绩，考试时间 100 分钟。

提示 1：在列方程前，先观察电路是否可以进行等效和化简

提示 2：大部分答案的数值都是简单整数或分数

I. (40 points) Answer the following questions based on the circuit shown below:



$R_1 = 3\,\mathrm{k}\Omega$ $C_1 = 1\,\mu\mathrm{F}$ $V_{S1} = 3\,\mathrm{V}$

$R_2 = 1\,\mathrm{k}\Omega$ $C_2 = 2\,\mu\mathrm{F}$ $V_{S2} = 6\,\mathrm{V}$

$R_3 = 1\,\mathrm{k}\Omega$ $L = 9\,\mathrm{m\,H}$ $I_{S1} = 2\,\mathrm{m\,A}$

$R_4 = 1\,\mathrm{k}\Omega$ $I_{S2} = 1\,\mathrm{m\,A}$

$R_5 = 2\,\mathrm{k}\Omega$

(1) How many KCL equations can be written for this circuit at most? How many independent KCL equations are there? How many KVL equations can be written using mesh analysis? (6 pts)

(2) Draw the equivalent circuit under DC steady-state conditions. (6 pts)

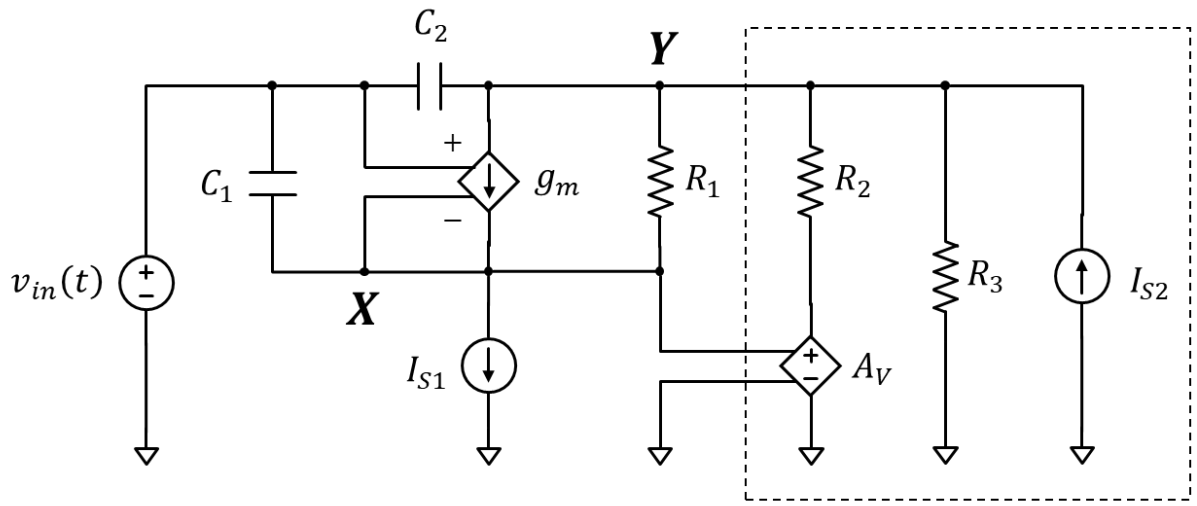
(3) Calculate the open-circuit voltage, short-circuit current, and DC equivalent resistance at port AB. (8 pts)

(4) Calculate the energy stored in each capacitor and inductor under DC steady-state conditions. (6 pts)

(5) Calculate the power of each source under DC steady-state conditions, and determine whether each is delivering or absorbing power. (6 pts)

(6) Assuming the external I-V characteristics at port AB remain unchanged, simplify the circuit so that it contains the minimum possible number of basic components (resistors, capacitors, inductors, and sources). (8 pts)

II. (40 points, 8 points per question) Answer the following questions based on the circuit shown below:



$R_1 = 1\text{k}\,\Omega$ $C_1 = 1\mu\,\text{F}$ $g_m = 1\text{m}\,\text{S}$ $I_{S1} = 1.5\text{m}\,\text{A}$

$R_2 = 3\text{k}\,\Omega$ $C_2 = 2\mu\,\text{F}$ $A_V = 3$ $I_{S2} = 2\text{m}\,\text{A}$

$R_3 = 1.5\text{k}\,\Omega$

(1) Find the Thevenin equivalent circuit for the circuit enclosed in the dashed box.

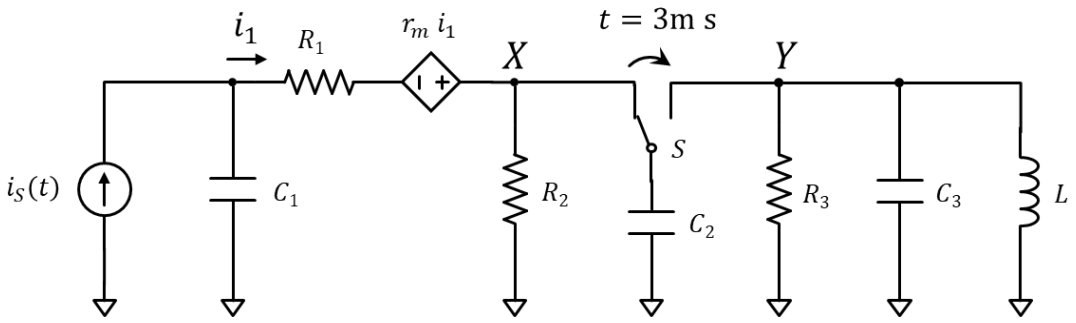
(2) Calculate the DC equivalent resistance from node X to ground.

(3) Assuming $v_{in}(t)$ varies very slowly, determine the relationship expressions for the node voltages $v_X(t)$ and $v_Y(t)$ in terms of $v_{in}(t)$.

(4) **Without the assumption** that $v_{in}(t)$ varies slowly, determine the generalized relationship expressions for the node voltages $v_X(t)$ and $v_Y(t)$ in terms of $v_{in}(t)$ after the zero-state transient response has settled (i.e., in steady state).

(5) Defining node X (with respect to ground) as Port 1, node Y (with respect to ground) as Port 2, and setting **all independent sources to zero**, determine the ***reciprocity*** and the ***active/passive nature*** of the circuit **under DC**.

III. (40 points, 8 points per question) Consider the circuit shown in the figure below. Assume that the circuit has reached stable for $t < 0$. $i_S(t)$ is a current source with step changes (its behavior is shown in the figure), and switch S changes its connection from the left contact to the right contact at time $t = 3\text{m s}$.

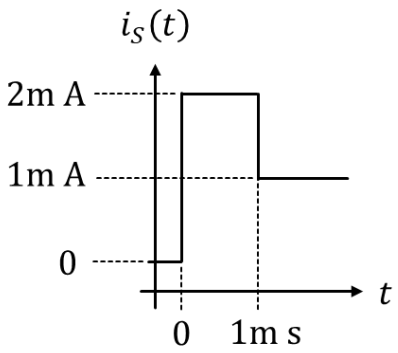


$R_1 = 1\text{k}\,\Omega$ $C_1 = 1.25\mu\text{ F}$

$R_2 = 1\text{k}\,\Omega$ $C_2 = 750\text{n F}$

$R_3 = 2\text{k}\,\Omega$ $C_3 = 250\text{n F}$

$r_m = 1\text{k}\,\Omega$ $L = 4\text{ H}$



Analyze:

(1) During the interval $0 \le t < 3\text{m s}$, is the voltage response at node X a first-order or second-order response? If it is a first-order response, calculate its time constant (τ); if it is a second-order response, calculate its damping ratio (ζ) and natural frequency (ω_0).

(2) Find the expression for the voltage response $v_X(t)$ at node X during the interval $0 \le t < 3\text{m s}$, and **sketch its waveform**.

(3) For the interval $t \ge 3\text{m s}$, is the voltage response at node Y a first-order or second-order response? If it is a first-order response, calculate its time constant (τ); if it is a second-order response, calculate its damping ratio (ζ) and natural frequency (ω_0).

(4) Find the expression for the voltage response $v_Y(t)$ at node Y for $t \ge 0$, and **sketch its waveform**.

(5) Assume that starting from $t < 0$, switch S continuously toggles between the left and right connections at a **very high** frequency. If the excitation current source $i_S(t)$ remains identical to the previous description, find the approximate response expressions for node X and node Y.